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Assessment of long-term sustainable end-use energy demand in Romania*

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ABSTRACT

Romania is the 10th largest economy in EU-28 and also one of the fastest growing economies in the region. An end-use energy demand model is developed for Romania to assess energy requirement by sector and by end-use for 2015–2050 period. Industry would surpass residential sector as the largest final energy-consuming sector from 2035 onwards. Services sector would exhibit the fastest growth of energy consumption. Despite expected decline in country's population, demand for electricity would grow in the future driven by increased household income and expanded services sector, which is relatively electricity intensive. Still, Romania's per capita electricity consumption would be about half of the EU-28 average. At the end-use level, thermal processes in industry, space heating in the residential and services, and road passenger travel in transport sector would be dominant throughout the study period. Improvement of energy efficiency in the heating system exhibits the highest potential of energy saving.

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KEYWORDS
End-use energy demand; energy modelling; energy-consuming sector; sustainability; Romania

1. Introduction

Romania, with a population of 19.7 million, is the 10th largest economy in the 28 member countries of the European Union (EU-28). After the collapse of oppressive regime in 1989 and its entry to the EU in 2007, Romania experienced an impressive economic growth over the past two decades. Except for the deep recession in 2009 and 2010, Romania's average annual gross domestic product (GDP) grew by 7% during 2003–2008 and by 2.2% during 2011–2015, well above the rate of EU-28 as a whole (Eurostat 2016a). Between 1995 and 2015, following the strong economic growth and declining number of people added to the population, country's per capita GDP increased by 94%, more than three-fold as fast as Germany (30%) and EU-28 average (32%). Despite the impressive economic growth, the current levels of per capita GDP and energy consumption in Romania are well below the EU average. For example, in 2014, GDP, final energy consumption and electricity consumption, in per capita terms, are 73%, 48% and 61% below the EU-28 average, respectively (Eurostat 2016a, 2016b).

As the Romanian economy is expected to expand and households' incomes and expenditure rise, more energy is needed to satisfy the growing demand. For example, the average monthly household expenditure on non-food goods, which currently represents 22% of the total expenditure, is increased by two-folds (8% per year) between 2005 and 2014 (INS 2016a). Likewise, the household ownership of refrigerator grew on average by 4.3% per year, followed by vacuum cleaner (2.3%), washing machine (1.9%), television (1.3%), gas cooking stoves (0.4%) and passenger car (0.4%) during 2008–2014 period (INS 2016b). If no action is taken, increasing energy demand will pose a challenge to Romania to comply its greenhouse gas (GHG) obligations set forth by the EU with its member countries.

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The EU sets number of short- and long-term quantitative climate and energy targets for its member countries, notably 20–20–20 climate and energy targets, 2030 framework for climate and energy policies and roadmap for moving to a low-carbon economy in 2050. As part of the broader EU energy and climate strategies and policies, Romania is committed to energy security, energy efficiency and competitive energy market improvement, renewable energy promotion and low-carbon green growth development. In particular, development of renewable energy in the country grew rapidly in recent years due to strategies and action plans such as green certificate system (Zamfir, Colesca, and Corbos 2016). In 2016, Ministry of Energy (MoE) published Energy Strategy of Romania for 2016–2030 with an outlook to 2050 that highlighted strategies for energy sector development (MoE 2016). A review study by Muresan and Attia (2017) analyse existing building energy efficiency in the country and discussed high-performance buildings in the future, such as Passive House, nearly Zero Energy Buildings and Net Zero Energy Buildings. Further, based on the assessment of 2014 national reform and convergence programmes for Romania, EU commission recommended energy programme that focuses on improving efficiency of industries, thermal insulation of buildings and the rehabilitation of district heating systems (EU 2014). There exists only a limited number of empirical studies that examine future energy demand in Romania. A recent World Bank report expects a significant increase in demand for final energy, particularly in transport and services sectors (World Bank 2014). Demand for electricity is also expected to rise in the future (Bianco et al. 2013). Using econometric techniques, (Bianco et al. 2010) estimate long-run GDP and price elasticities of non-residential electricity consumption in Romania and finds that these elasticity values are quite low, i.e. 0.496 (income elasticity) and −0.274 (price elasticity). The sixth National Communication of Romania to the United Nations Framework Convention on Climate Change (MECC 2013) projects that the country's demand for final energy would increase by 28% between 2010 and 2030, and that residential and industry sectors are responsible for over 60% of this increase. Likewise, using PRIMES model, EU (2016) projects country's final energy to grow by 15% between 2010 and 2050, with three-fourth of this increase is coming from industry and transport sectors. Analysing historical final consumption of energy over the past decade, ICEMENERG and ANRE (2012) find that country's demand for energy in industry sector is declined, mainly due to shifting away from energy-intensive manufacturing industries, while it is increased in rest of the sectors, thereby confirming a structural change of the economy. A study by Grafakos, Enseñado, and Flamos (2017) developed a framework of sustainability and resilience indicators for assessing low-carbon energy technologies in Europe. Based on energy policy and legislation, Colesca and Ciocoiu (2013) presented Romanian energy system with main focus on renewable energy sources. However, these studies lack analysis of detailed long-term projections of energy demand at sub-sector and end-use levels. Information regarding how energy is used across various end-uses in the future is critical for formulating energy plans and policies.

The paper makes three main contributions. First, we develop flexible and easy to learn end-use bottom-up model to study long-term energy demand at sub-sector and end-use level for Romania. While there are long established energy system models, our model requires less significant learning curve and time commitment, less input data requirement and non-proprietary software to assess energy demand. Second, we analyse detailed long-term energy demand evolution by sub-sector and by end-use for Romania. This is especially important because understanding of these detailed energy demand evolution is critical in implementing EU recommended energy and climate strategies in the country. Our third contribution is identifying the key sector and sub-sector strategies for the development of sustainable energy and low-carbon economy over the long-term in Romania.

2. Overview of current energy demand and economy in Romania

Romania's final energy consumption is in downward trend over the past decade. In 2014, country's total final consumption (TFC) of energy, which is the sum of consumption by different end-user sectors, is 21.7 million ton of oil equivalent (Mtoe), about 13% lower than in 2004. This is mainly due to structural changes and declining energy intensity in the industry sector. Between 2004 and 2014, the industry and residential sectors cut consumption by 35% and 7%, respectively, while the consumption is increased in transport (19%) and services (40%) sectors. Residential, industry and transport sectors are the major end-users of energy in the country, representing combined 90% of TFC in 2014. Since 2009, the residential sector surpassed the industry sector as the country's largest consumer of TFC. The high share of country's residential sector in TFC in 2014 (34%) is also the second greatest energy consumer in all EU-28 countries after Croatia (Eurostat 2016b), reflecting high heating demands in cold climate and an ageing housing stock. The industry sector including agriculture (33%), transport sector (25%) and the services sector (8%) are the second, third and fourth largest consumers of TFC in the country. By fuel types, oil products are the largest fuel consumed, accounting for 31% of TFC in 2014, with more than 77% consumed by the transport sector. Natural gas, mainly used in industry and residential sectors, accounted for 26% of TFC, while biomass and renewable wastes combined accounted for 17%, with more than 86% used in residential sector. The remaining 26% of TFC comes from electricity (17%), district heat (6%) and solid fuels (mainly coal) (3%).

In terms of purchasing power parity (PPP), Romania's economy ranks quite high within the EU-28 (10th). For example, country's GDP reached US$ 380 billion in 2014 (measured in 2011 US$), which is almost doubled from 1992 level (World Bank 2016). In per capita terms, country's real GDP is increasing steadily over the last two decades. Between 1992 and 2014, Romania's real per capita GDP is more than doubled, grew at much faster rate (3.5% per year) than in Germany (1.2%) and the EU-28 as a whole (1.5%) (Table 1). However, the current level of country's real per capita GDP remains much lower in the region, e.g. it is less than half of Germany level and a little more than half the level of EU-28 as a whole. Further, per capita final household expenditure of Romanian households grew significantly by an average rate of 5.4% per year between 1992 and

Table 1. Historical socio-economic and TFC of energy indicators in Romania, Germany and EU-28[a].

	Romania			Germany			EU-28		
	1992	2014	% change 1992–2014	1992	2014	% change 1992–2014	1992	2014	% change 1992–2014
GDP per capita (PPP, constant 2011 US$)	9042	19,104	111%	33,215	43,559	31%	25,110	34,796	39%
Household final consumption expenditure per capita (PPP, constant 2011 US$)	3315	10,272	210%	18,767	23,046	23%	13,657	18,470	35%
Electricity consumption per capita (kWh[b])	1816	2101	16%	5617	6350	13%	4547	5338	17%
TFC of energy per capita (kgoe[c])	1192	1088	−9%	2753	2586	−6%	2224	2093	−6%
Industry, TFC (% of total)	52	30	−43%	29	29	1%	31	26	−16%
Residential, TFC (% of total)	23	34	48%	28	25	−12%	27	25	−6%
Transport, TFC (% of total)	15	25	73%	28	30	9%	28	33	19%
Services, TFC (% of total)	2	8	252%	13	16	23%	11	13	25%
Agriculture, TFC (% of total)	8	3	−65%	2	0	−98%	4	3	−33%
Total, TFC intensity (PPP, kgoe per thousand 2011 US$)	132	57	−57%	83	59	−28%	88	60	−32%
Industry, TFC intensity (PPP, kgoe per thousand 2011 US$)	156	62	−60%	66	57	−14%	89	64	−27%
Services, TFC intensity (PPP, kgoe per thousand 2011 US$)	8	7	−17%	17	14	−19%	14	11	−23%
Energy dependence[d] (%)	30	17	−43%	55	61	13%	46	53	17%

[a]For comparison purposes, we arbitrarily chose Germany, which represents one the most advanced country in region, and the EU-28 average.
[b]kWh is kilo-watt hour.
[c]kgoe is kilogram of oil equivalent.
[d]Energy dependence is calculated as net imports divided by the sum of gross inland energy consumption plus bunkers.
Sources: Eurostat (2016b); World Bank (2016); EU (2015).

2014. This growth rate of Romania's per capita final household expenditure is highest in the EU-28 countries but its current level, in absolute term, remains much lower compared to the level of Germany and EU-28 average.

Likewise, country's electricity and TFC, in per capita terms, are still much lower compared to the respective average values for EU-28 and Germany. For example, in 2014, Romanians consumed about one-third of electricity to that of Germans and about 39% to that of EU-28 average (Table 1). Similarly, country's per capita TFC is less than half (42%) to that of Germany and 52% to that of EU-28 average in the same year. Over the last decades, Romania's primary production and net import of fossil fuels (coal, oil and gas) continues to decline, while primary production of nuclear heat and renewables continues to increase. Overall, country's gross inland consumption of all fuels declined more than the net import (in absolute terms). This led to decline in energy dependency, which shows the extent to which an economy relies upon imports in order to meets its energy needs, from 30% in 1992 to 17% in 2014 (Table 1).

There is strong correlation between the levels of sectoral TFC and income. For instance, TFC of both residential and transport sectors and income in Romania is much lower compared to Germany and the EU-28 average during 1992–2014 period (Figure 1, top). Similarly, there is a close correlation between income and sectoral final energy intensity. In particular, final energy intensity of industry sector relative to GDP per capita is declining in Romania, Germany and the EU-28 average during 1992–2014 period (Figure 1, bottom). In contrast, correlation between energy intensity of services

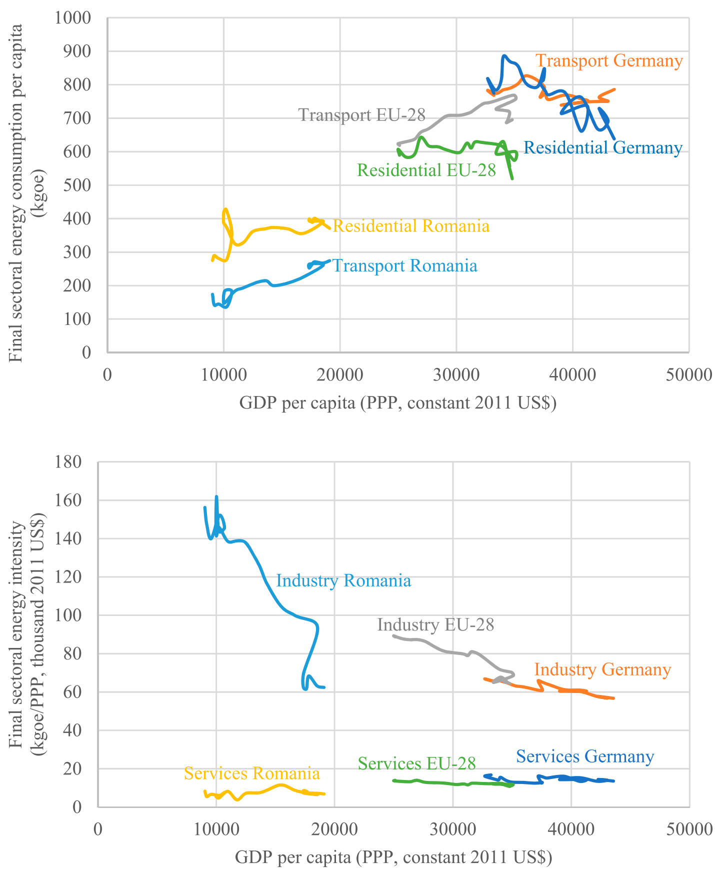

Figure 1. Relation between income and final sectoral energy demand in per capita term (top), and income per capita and final sectoral energy intensity (bottom) in Romania, Germany and EU-28 during 1992–2014 period. Sources: Eurostat (2016b); World Bank (2016).

sector relative to GDP per capita is about the same during the same period. However, these sectoral intensities are relatively much higher in Romania when compared with high-income countries like Germany and the EU-28 average. More specifically, average final energy intensity of industry in the country declined significantly by 70% during the same period.

3. Methodology

In the literature, various modelling techniques are used for analysing end-use energy demand of different energy sectors of the economy. These techniques vary from a complex mathematical programming with extensive input data to a simple modelling framework with limited input data.[1] In general, two types of approaches, econometric and end-use accounting, are commonly used in the existing energy demand models in developing countries (Swan and Ugursal 2009; Bhattacharyya and Timilsina 2010; Suganthi and Samuel 2012). The first approach is often used at aggregated level such as total energy demand. In this approach statistical relationship between energy consumption and macroeconomic variables, such as GDP, is established based on historical data and the same relationship is used to forecast future energy demand. Such an approach is not applicable when detailed energy demand forecasting is needed at the end-use level because historical data on detailed end-use energy consumption is not available. An end-use accounting approach, which does not need time series data, but rely on detailed data for a reference year, is employed for forecasting end-use energy demand at various sectoral levels. For this study, we developed an end-use energy demand analysis (EEDA)[2] model for Romania. The EEDA is a simple bottom-up accounting framework model for evaluating long-term end-use service demand at sub-sector and end-use levels in Romania. For an end-use in a given sector, the main elements of energy demand in our model are activity, structure and intensity. The evolution of socio-economic and technological factors are the main driving forces of future energy demand.

Four energy-consuming sectors (residential, services, industry and transport) are considered in the model. The services sector is further divided into seven sub-sectors by type of buildings: office, educational building, hospital, hotel and restaurant, sport facilities, wholesale and retail store, and others (not classified elsewhere). Industry sector is divided into four sub-sectors: agriculture, construction, mining and quarrying, and manufacturing. The manufacturing sub-sector is further sub-divided into 10 manufacturing industry types based on economic activities in the European Community classification (EC 2008). Transport sector is further sub-divided into five sub-sectors (road, rail, air, inland waterways, maritime and pipeline) based on mode of transportation. Residential sector could be divided into sub-sectors by geography (e.g. rural and urban) or income level (e.g. low, medium and high). However, due to lack of reliable data, we do not classify residential sub-sectors. A simplified classification of the sector, sub-sector and end-use of the model is presented in Figure 2.

The energy demand for each end-use categories is driven by one or several demographic, socio-economic and technological parameters, whose values are given as part of the scenarios. Demand of energy use by each sub-sector and by each end-use is calculated using the following general equations:

$$E_X(t) = \sum_x E_{x,X}(t) = \sum_x \sum_y E_{y,x,X}(t), \quad x \in X, \tag{1}$$

$$E_{y,x}(t) = A_{y,x}(t) \cdot S_{y,x}(t) \cdot EI_{y,x}(t), \tag{2}$$

where E_X is the total energy demand for sector X, A is the activity level, S is the structure and EI is the energy intensity. The small cap subscripts x and y represent sub-sector and end-use, respectively. For example, residential sector end-uses include space heating, water heating, air conditioning, cooking, lighting and use of electric appliances. Activity level may be value-added or domestic production for industry sector, domestic demand and commercial floor area for services sector, population, living

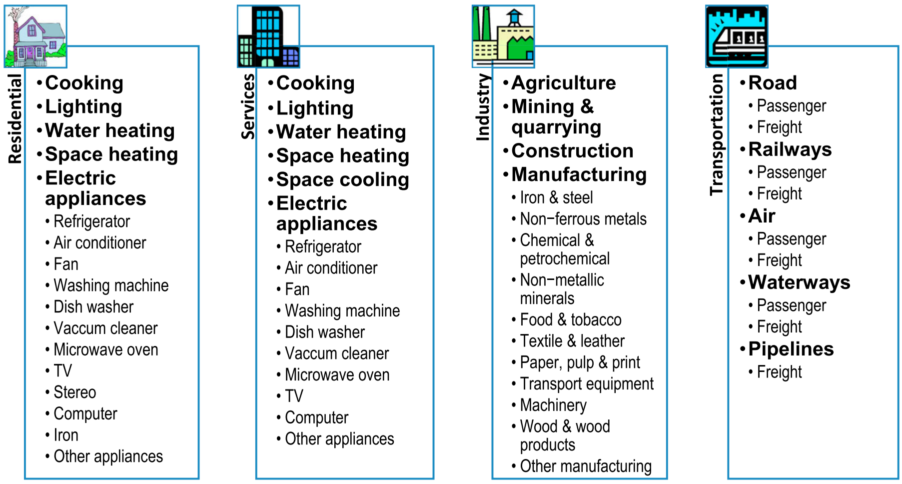

Figure 2. Simplified classification of sector, sub-sector and end-use classification adopted in EEDA model for Romania.

floor area and ownership of electrical appliances for residential sector, and passenger-kilometre (pkm) or ton-kilometre (tkm) for transport sector. Likewise, sectoral structure is the mix of activities within a sector and sub-sectors, such as the share of domestic production in manufacturing industries, and energy intensity is the energy use per unit of activity, such as energy use per domestic production for manufacturing industries. Depending on data availability, breakdown of energy demand by sub-sectors and end-uses for each sector are different. Note that demand for energy is not disaggregated by fuel types. The reason for this is the fuel-mix largely depends on the technological possibilities of supply and their relative prices, which is outside the scope of this paper. However, due to its non-substitutability nature and importance, the demand for electricity by sector is calculated separately in TFC.

In addition to estimating transport energy demand, activity–structure–intensity–fuel (ASIF) framework approach is used for calculating travel demand. The following equations are used:

$$TD_{k,T}(t) = \sum_j TD_{j,k,T}(t) = \sum_j TDD_{j,k,T}(t) \cdot TI_{j,k,T}(t), \quad j \in T, \tag{3}$$

$$TI_{j,k,T}(t) = TI_{j,k,T}(t-n) \cdot (1 + R_{j,k,T}(t))^n, \tag{4}$$

$$R_{j,k,T}(t) = TE_{j,k,T}(t) + PE_{j,k,T}(t) + RE_{j,k,T}(t), \tag{5}$$

where *TD* is travel demand, *TDD* is transport domestic demand (activity variable), *TI* is transport mobility intensity, *R* is growth rate of overall change in mobility intensity, *TE* is growth rate of travel efficiency improvement change, *PE* is growth rate of population change, *RE* is growth rate of rebound effect change and *T* is transport sector, *t* is time period and *n* is number of period. The small cap subscripts *j* and *k* represent transport mode (e.g. air, road, rail, water or pipeline) and activity type (passenger or freight), respectively. In this framework, the overall level of travel demand depends on transportation mode, mobility intensity, travel efficiency, population change and rebound effect. In this analysis, travel demand for all modes of transportation is calculated as a function of historical value of travel demand and corresponding driving variables. Mobility intensity is influenced by travel efficiency, change in population and rebound effect. In the analysis, it is also hypothesised that rebound effect has positive impact on private passenger transportation (0.6–1.0% per year) and negative impact on public passenger transportation (0.15–0.3% per year). Passenger travel demand is measured in pkm or in number of passengers, and freight travel demand is measured in tkm or in ton. In Romania, land transport includes road, rail and pipeline, water transport includes maritime and inland waterways, and air transport includes aircrafts. Excluding pipeline, each of these transport modes are further sub-divided by activity level (e.g. passenger and freight). Since road dominates Romania's transportation system, it is further sub-divided by private and public road transport.

4. Data, scenario description and key assumptions

The data used in the study are grouped under three categories: energy, socio-economic and demographic, and technological data. The primary source of these data are Romania's National Institute of Statistics (INS), EU's Eurostat, the World Bank's World Development Indicators, International Energy Institute (IEA)'s energy balances, and Building Performance Institute (BPIE)'s buildings data. Since none of these publications provide complete dataset, additional dataset are compiled from Ministry of Environment and Climate Change (MMSC)'s sixth National Communication on Climate Change report (MECC 2013), International Telecommunication Union (ITU)'s Measuring the Information Society 2014 report (ITU 2014), ENTRANZE project report (Atanasiu, Economidou, and Maio 2012) and EU's Joint Research Centre (JRC) Scientific and Policy reports (Bertoldi, Hirl, and Labanca 2012; Pardo et al. 2012), in particular, for calibrating base year and activity parameters used in projecting energy demand.

The starting year of the analysis is 2013 (base year). The projection is made through the year 2050 with five years of interval starting from 2015. Three scenarios are considered: baseline, low demand and high demand scenarios. These scenarios are differentiated primarily by their underlying assumptions about socio-economic and technological factors. The baseline scenario takes into account the current trends on socio-economic development, sectoral energy-use patterns and technological progress. It reflects the path of future energy demand given a continuation of current trends and policies.

To illustrate increase in end-use energy demand (optimistic pathway) with respect to baseline scenario, a high scenario is constructed. This scenario is characterised by the Romanian economy and population growing faster than the baseline scenario. This scenario also reflects households with high home electric appliances and private vehicle ownership, and more passenger and freight transport mobility. Overall, in this scenario, the general development mode of Romanian economy is optimistic. In contrast, to illustrate decrease in end-use energy demand (pessimistic pathway) with respect to baseline scenario, a low scenario is constructed. In this low scenario, economic and population growth are slightly lower compared to baseline scenario. Due to slower economic growth, domestic production and domestic demand are also lower. The selected key driving variables under the baseline and two alternative scenarios and assumptions are summarised in Appendices 1 and 2.

5. Results and discussion[3]

5.1. TFC of electric and non-electric energy at the national level

From 2013 to 2050, TFC of electricity in the baseline is projected to increase from 3493 kilo ton of oil equivalent (ktoe) to 5668 ktoe, at an average rate of 1.3% per year (Table 2). Industry remains the single largest electricity user, accounting for almost half of total electricity demand over the projected period. By 2050, TFC of electricity in services, one of the fastest growing sectors in the country, is projected to be about the same as that of residential sector. Despite transport's share in TFC of electricity is small (3.5%) in 2050, this sector exhibits the fastest rate of expansion in percentage terms, at an average rate of 2% per year, primarily due to increasing use of electric vehicles.

However, demand for electricity is markedly different across the scenarios. In high scenario, which reflects optimistic socio-economic development in the country, TFC of electricity is projected

Table 2. TFC of electricity by sector in the baseline scenario, 2013–2050 (ktoe).

	2013	2015	2020	2025	2030	2035	2040	2045	2050
Industry	1689	1703	1869	2060	2234	2393	2570	2721	2867
Residential	1023	1035	1086	1132	1170	1211	1258	1303	1325
Services	685	719	810	880	954	1035	1119	1196	1276
Transport	96	100	110	122	135	149	164	181	200
Total	3493	3557	3875	4194	4493	4788	5111	5401	5668

to grow on average by 1.8% per year during 2013–2050, reaching almost 6681 ktoe in 2050, an increase of 91% from 2013 value. Electricity demand expands much more rapidly in high scenario compared to baseline to a level in 2050 that is 18% higher (Figure 3). In contrast, electricity demand expands much slower in low scenario, which reflects pessimistic socio-economic development in the country, at an average rate of 0.9% per year to a level in 2050 that is 13% lower from the baseline.

As a result of rising demand for electricity and declining population in the country during 2013–2050 period, per capita TFC of electricity is projected to increase. For example, in the baseline, it is projected to increase from 2033 kWh in 2013 to 3698 kWh in 2050, an increase of 82% (Figure 4). However, the differences in per capita TFC of electricity in Romania and other advanced countries in the EU remain very large. In 2040, per capita TFC of electricity in Romania amounts to 3239 kWh, while it is 5710 kWh in Germany and 6294 kWh in EU-28 (average). Note that Romania's projected per capita TFC of electricity in 2050 is much lower than that of Germany and average EU-28 in 2013. As a reference, if electricity use by each Romanian increased to the 2013 level of German, Romania's electricity use would increase by three-folds.

As economy grows, Romania's TFC of non-electric energy is also projected to increase. Between 2013 and 2050, non-electric energy demand in the baseline is projected to increase on average by 0.7% per year, reaching 23,125 ktoe in 2050 (Table 3). The share of industry in TFC of non-electric energy, increases gradually, from 28% in 2013 to 33% in 2050. In contrast, the residential sector's share is projected to slightly decline over the same period, at an average rate of 0.05% per year, mainly due to declining population. Between now and 2030, residential sector is projected to be the largest end-user of TFC of non-electric energy. However, from 2035 to 2050, industry sector surpasses residential sector as the largest TFC of non-electric energy end-user, mainly due to strong growth in overall industrial production. Despite declining population and improvement in vehicle fuel-economy during the projected period, rising number of motor vehicles and increasing economic activities would lead to rise in TFC of non-electric energy in transport sector. For example, transport's TFC of non-electric energy is projected to increase steadily, from 5182 ktoe in 2013 to

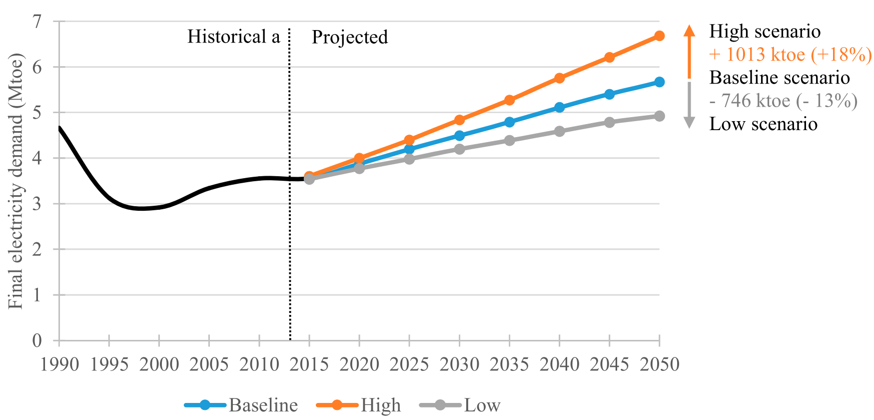

Figure 3. Historical and projected TFC of electricity by scenario in Romania, 1990–2050.
[a]Historical data are taken from INS (2016c).

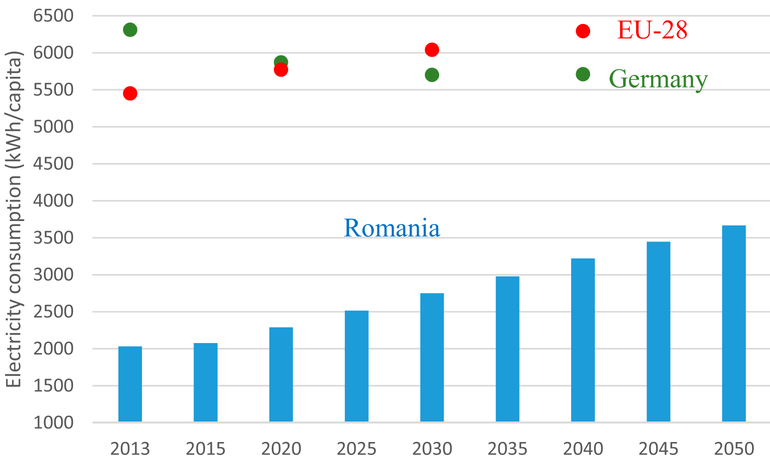

Figure 4. TFC of electricity per capita in the baseline scenario, 2013–2050.
Note: Data for EU-28 and Germany are taken from Eurostat (2016b); IEA (2013, 2014).

6793 ktoe in 2050, an increase of annual average of 0.73%. Although services sector's share in TFC of non-electric energy is small (13%) in 2050, this sector would exhibit the fastest rate of growth in percentage terms, at an average rate of 1.7% per year, reflecting the structural change in the country from manufacturing to services sector.

The magnitude and growth rate patterns of TFC of non-electric energy are quite different across the scenarios. In high scenario, the changes in industry's TFC of non-electric energy compared to baseline, is projected to increase at much higher percentage rate than in other energy-consuming sectors. For example, compared with baseline scenario, the amount of TFC of non-electric energy in industry ranges from 5% higher in 2020 to 23% higher in 2050 in the high scenario. Likewise, it ranges from 4% to 20% higher in services sector, 2% to 8% higher in residential sector and 1% to 7% higher in transport sector over the same time period (Figure 5). In contrast, changes in services sector's TFC of non-electric energy in the low scenario when compared with baseline scenario is projected to decrease at much higher rate than in other sectors. For example, the thermal energy demand in services decreases from 4% in 2020 to 17% in 2050, while it decreases from 3% to 12% in industry, 1% to 9% in residential and 1% to 4% in transport over the same time period.

Although comparing energy demand forecast between studies is not straightforward, due to different methodologies and assumptions, we compare our study with selected few relevant studies.[4] For example, the EU estimate of TFC of all energy (electric and non-electric) in 2050 is 27.3 Mtoe compared to our estimate of 28.8 Mtoe. Likewise, EU estimate of TFC of electricity in 2050 is 5.7 Mtoe compared to our study estimate of 5.8 Mtoe. The possible reason for slightly lower EU estimate of TFC compared to our study is the assumption of country's lower economic growth in the EU study. Further, the Ministry of Environment and Climate Change (MECC) estimate of TFC of energy without any measure (28 Mtoe) in 2030, the latest year reported, is slightly higher compared to our

Table 3. TFC of non-electric energy[a] by sector in the baseline scenario, 2013–2050 (ktoe).

	2013	2015	2020	2025	2030	2035	2040	2045	2050
Industry	5078	5118	5507	5941	6353	6723	7109	7397	7669
Residential	6699	6691	6716	6716	6676	6659	6667	6670	6586
Services	1101	1164	1316	1427	1541	1677	1818	1946	2077
Transport	5182	5279	5531	5740	5957	6199	6452	6620	6793
Total	18060	18252	19070	19824	20527	21258	22046	22633	23125

[a]TFC of non-electric energy includes coal, oil, gas, heat and renewables combined.

Figure 5. Changes in TFC of non-electric energy by sector in high and low scenario compared to baseline scenario.

study in the baseline (25 Mtoe), possibly due to higher economic growth assumption. The National Commission for Prognosis (NCP) made projection of TFC of energy by sector up to year 2018. For comparison, the NCPs estimates of TFC of energy for residential (7.8 Mtoe), industry (7.1 Mtoe) and transport (5.5 Mtoe) sectors in 2015 is similar to our study in magnitude. These comparisons suggest some similarities in the magnitude of TFC of energy in Romania.

5.2. TFC of energy at the sector level

5.2.1. Residential sector

The residential sector is Romania's largest energy consumer. Romanian households spend more than 13% of their income on energy, one of the highest rates in the EU countries (EU 2014). In the baseline, residential TFC of energy is projected to grow on average by 0.1% per year from 2013 to 2050 to reach 7911 ktoe (Table 4). The residential TFC vary markedly in the alternative scenarios. For example, it is projected to increase from 7722 ktoe in 2013 to 8547 ktoe in 2050 in high scenario, at an average of 0.3% per year. In contrast, it is projected to decrease (0.2% per year) in low scenario, to reach 7202 ktoe in 2050. Non-electric energy, mainly used for space and water heating by the households, contributes the most in TFC of residential energy, while the contribution of electricity,

Table 4. Residential end-uses TFC of energy by scenario, 2013–2050 (ktoe).

		Baseline			High			Low		
	2013	2020	2030	2050	2020	2030	2050	2020	2030	2050
Space heating[a]	4638	4707	4777	4861	4814	5023	5380	4624	4551	4310
Air conditioning	31	42	59	89	42	59	90	42	58	86
Water heating	1299	1267	1201	1094	1269	1207	1113	1266	1193	1060
Cooking	790	772	731	667	772	735	679	771	727	646
Lighting	218	223	231	244	228	243	270	219	220	217
Electric appliances	746	791	849	955	801	873	1014	782	823	884
Refrigerator/freezer	240	248	256	279	257	278	325	239	235	230
Washing machine	118	120	122	127	120	123	129	120	122	122
Television	176	184	194	210	184	195	214	184	193	202
Computer	28	39	53	79	39	54	81	39	53	76
Other appliances	184	200	223	261	201	224	266	200	221	252
Total	7722	7802	7847	7911	7926	8140	8547	7704	7573	7202

[a]Climate corrected.

mainly used for running electric appliances and lighting, is relatively small in TFC of residential energy in all three scenarios.

At the end-use level, space heating (climate corrected) is by far the largest energy consumer across all scenarios. In the baseline, space heating accounted for 60% of TFC of residential energy in 2013 and it is projected to remain about the same during 2015–2050 period (Figure 6). In contrast, energy demand for water heating and cooking is projected to decline slightly both in absolute and percentage terms during 2015–2050 period. For example, energy demand for these two end-uses is projected to decline on average by 0.5% per year in the baseline (Table 4). This is mainly due to country's declining population and improvement in efficiency of heating and cooking devices during the projected period. It follows the similar trend in alternative scenarios, with increasing in magnitude in high scenario and decreasing in magnitude in low scenario compared to the baseline.

Space heating, followed by water heating and cooking, are the three major end-use services in the residential sector. The highest share of energy requirements for space heating in the country is mainly due to long cold winter with higher heating degree days. Apart from outdoor temperature, many other factors influence energy demand for space heating, including the size and type of dwellings, and the efficiency of the heating system and equipment.[5] Space heating, therefore, represents the largest opportunity to reduce residential energy demand, for example, by using more efficient heating equipment and by changing energy-mix. For space heating, natural gas and derived heat are commonly used in urban households, while solid biomass fuels (including charcoal) are commonly used in rural households. In 2013, about half of total energy used for residential space heating came from solid biomass, followed by natural gas (31%) and derived heat (18%) (Eurostat 2016b). In the same year (2013), 80% of total residential energy is used for heating (space and water heating), 7% is used for cooking and the remaining 13% is used lighting and electric appliances including air conditioning.

As economy grows and income rises, the use of household electrical appliances and corresponding demand for electricity are also projected to increase. For example, demand for electricity by electric appliances is projected to increase from 746 ktoe in 2013 to 955 ktoe in 2050, at an average rate of 0.7% per year in the baseline. In particular, residential use of refrigerator, washing machine and television combined accounts for 72% of TFC of residential electricity by electric appliances during 2015–2050 period. Electricity demand for lighting is projected to increase at slightly lower rate of 0.3% per year to reach 244 ktoe in 2050. Despite air conditioning's share in total residential energy

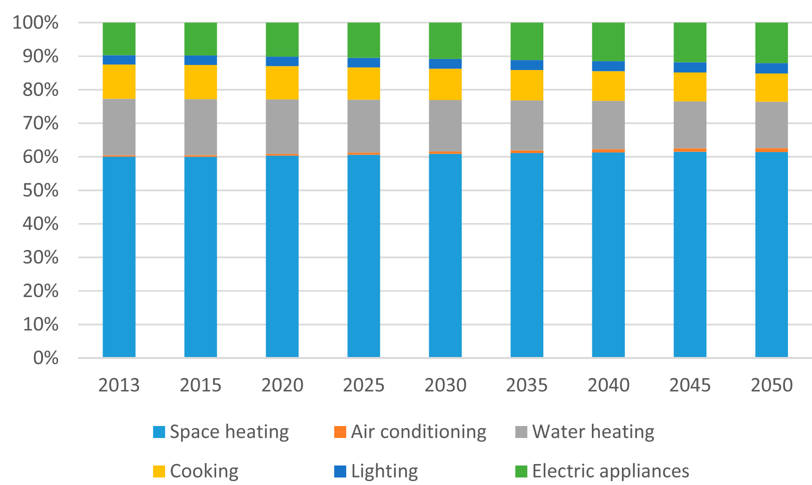

Figure 6. Percentage share of end-uses in total residential TFC of energy in the baseline scenario.

demand is small (1%) in 2050, this residential end-use would be the fastest rate of growth in percentage terms, at an average rate of 2.9% per year in the baseline.

5.2.2. Services sector

The importance of the services sector for Romania's energy policy is grown significantly over the past decade. In 2013, services sector accounted for more than half of country's GDP and about 8% of TFC of energy (INS 2016c, 2016d). Since 2002, energy demand increased by 41% as a result of an average growth of 3.5% per year. This sector is also the most heterogeneous sector of the economy that includes wide range of energy consumers. TFC of energy by services is projected to grow at much higher rate than other energy-consuming sectors in all three scenarios. For example, services sector's TFC of energy is projected to double from 1785 ktoe in 2013 to 3353 ktoe in 2050, at an average rate of 1.7% per year in the baseline (Table 5). Relative to baseline, it follows higher trends in high scenario and lower trends in low scenario.

At the end-use level, similar to residential sector, services sector space heating is projected to account for most of the energy demand. In 2050, the share of space heating in services TFC of energy accounts for 48% in the baseline, followed by others (25%), lighting (13%), water heating (8%) and air cooling (6%). At the sub-sector level, excluding energy demand for others, wholesale and retail store, and office buildings combined account for more than half of services TFC in all three scenarios during the projected period (Figure 7). Hotel and restaurant, hospital, educational building, sports facility and n.e.c. (not elsewhere classified) sub-sectors account for remaining half of the services energy demand in descending order during the projected period.

Likewise, in 2050, both at end-use and sub-sector level, the share of space heating is projected to account for the highest (37%) in wholesale and retail store to the lowest (4%) in n.e.c. sub-sector (Figure 8). In the case of lighting, it accounts for the highest (24%) in hospital to the lowest (4%) in sport facility sub-sector in the baseline. This is followed by water heating, in the range of 4% in n.e.c. to 24% in hotel and restaurant, and space cooling, in the range of 3% in n.e.c. to 29% in offices in 2050.

5.2.3. Industry sector

Despite a sharp decline in industry's TFC of energy since 1990, Romania remains heavily dependent on energy-intensive manufacturing industries. In 2013, about 87% of industry's TFC is consumed by manufacturing industries. In the same year, close to 70% of total manufacturing TFC is consumed by handful of energy-intensive industries, such as iron and steel, chemical and petrochemical, and non-metallic minerals (Eurostat 2016b). In terms of energy types, coke, natural gas and electricity are mainly used in iron and steel industry, while natural gas, electricity and refinery gas is mainly used in chemical and petrochemical industries, and petroleum coke, natural gas and electricity is mainly used in non-metallic minerals industries.

In the baseline, TFC of energy in industry is projected to increase from 6767 ktoe in 2013 to 10,536 ktoe in 2050, at an average rate of 1.2% per year (Table 6). Demand for electricity in this

Table 5. Services end-uses TFC of energy by scenario, 2013–2050 (ktoe).

		Baseline			High			Low		
	2013	2020	2030	2050	2020	2030	2050	2020	2030	2050
Space heating	840	1006	1179	1594	1046	1287	1919	968	1080	1322
Space cooling	94	113	135	188	117	147	227	108	123	156
Water heating	149	178	208	280	185	227	338	171	191	233
Lighting	231	273	326	437	284	356	527	263	299	363
Others[a]	473	556	647	853	578	706	1027	535	592	707
Total	1785	2126	2495	3353	2210	2724	4038	2045	2285	2781

[a]Others include energy demand for cooking (mainly in hotel/restaurant) and electricity for appliances and lighting in public places.

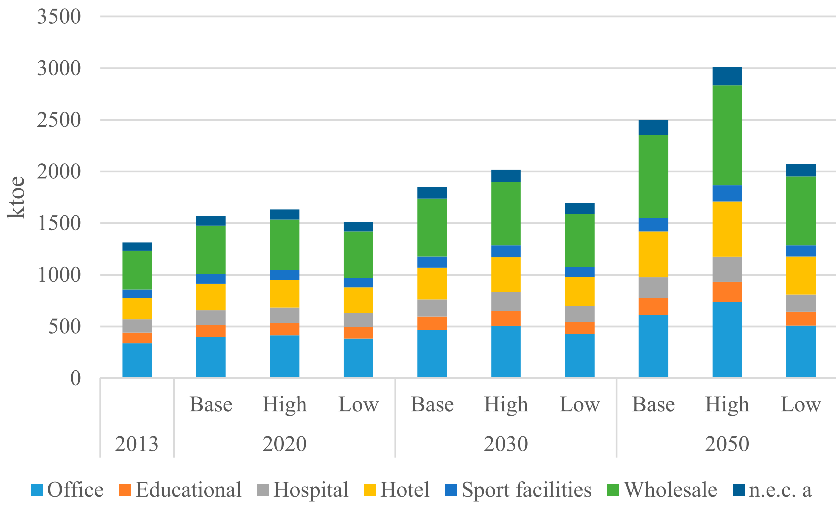

Figure 7. Services sub-sector TFC of energy by scenario, 2013–2050.
[a]n.e.c. include institutional buildings, warehouses and other non-specified service industries.

sector is projected to increase at much higher rate (1.4% per year) than the demand for non-electric energy (1.1% per year) in the baseline during 2015–2050 period. Note that electricity is used for only electric motors and others, while non-electric (thermal) energy is used for motive, thermal-electric and heat requirements. Relative to baseline, the demand for industrial energy is projected to increase in high scenario and decrease in low scenario.

At the end-use level, thermal (heat) energy use is projected to account for most of the TFC of energy in industry during the projected period in all three scenarios. For example, in 2050, the share of thermal energy in TFC accounts for 64% in the baseline. This is followed by electric motor (16%), others (11%), and motive and thermal-electric power (9%) (Figure 9, right). At the sub-sector level, manufacturing industry is by far the largest end-user in terms of total industry energy use. In 2050, manufacturing industry is projected to account for 86% of total industry energy

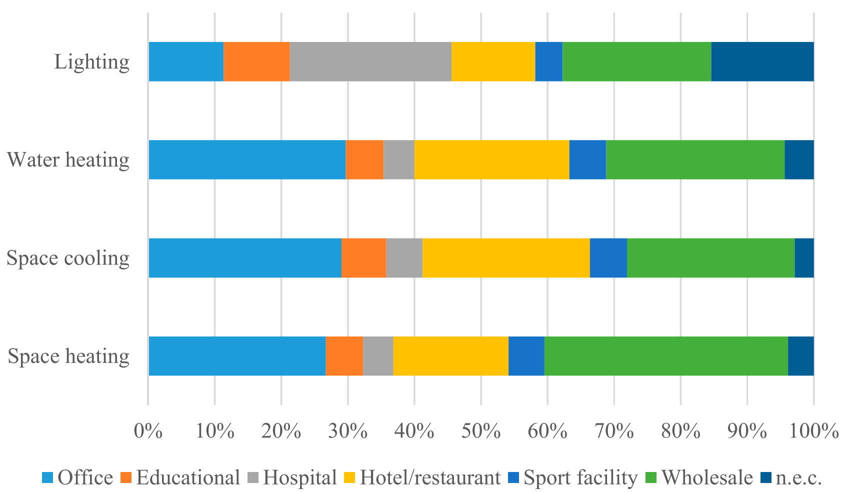

Figure 8. Services end-use and sub-sector TFC of energy in the baseline scenario, 2050.

Table 6. Industry sub-sector and end-use TFC of energy by scenario, 2013–2050 (ktoe).

		Baseline			High			Low		
	2013	2020	2030	2050	2020	2030	2050	2020	2030	2050
Electric motor	1013	1121	1341	1720	1165	1465	2094	1088	1241	1487
Agriculture	42	45	48	51	46	52	61	43	44	43
Manufacturing	932	1036	1240	1596	1078	1357	1947	1006	1149	1383
Mining and quarrying	12	12	13	14	12	14	16	12	12	11
Construction	27	28	40	59	29	42	70	27	36	49
Others (electricity)	676	748	894	1147	777	977	1396	725	827	991
Agriculture	28	30	32	34	31	35	41	29	29	28
Manufacturing	621	691	827	1064	718	904	1298	671	766	922
Mining and quarrying	8	8	9	10	8	9	11	8	8	8
Construction	18	19	26	39	19	28	46	18	24	33
Motive/thermo-electric	537	577	707	916	592	761	1088	553	644	767
Agriculture	253	268	294	316	277	319	378	258	269	265
Manufacturing	71	85	104	139	88	113	167	82	95	119
Mining and quarrying	20	20	22	25	21	24	29	20	20	20
Construction	193	203	286	436	207	307	514	193	260	364
Thermal energy	4540	4817	5382	6163	5029	5951	7633	4696	5038	5440
Agriculture	131	132	136	130	136	148	155	127	125	109
Manufacturing	4251	4634	5286	6303	4857	5867	7833	4537	4969	5586
Mining and quarrying	3	3	3	3	3	3	4	3	3	3
Construction	156	161	220	316	164	236	373	153	200	264
Total (industry)	6767	7376	8587	10536	7695	9457	12942	7185	8009	9206

demand, followed by construction (8%), agriculture (5%) and mining and quarrying (1%) in the baseline (Figure 9, left).

In manufacturing industries, iron and steel, chemical and petrochemical, and non-metallic minerals combined is projected to account for two-thirds of total manufacturing energy demand over the study period in the baseline (Figure 10). The other notable manufacture sub-sectors include machinery, and food and tobacco. The share of these two sub-sectors combined in total manufacturing energy demand is projected to increase from 15% in 2013 to 22% in 2050 in the baseline. In absolute values, the top three energy-consuming manufacturing sub-sectors are chemical and petrochemical (3072 ktoe), iron and steel (1424 ktoe) machinery (1052 ktoe) and non-metallic minerals (1015 ktoe) in 2050.

In industry sector, three temperature intervals are considered to describe the quality of thermal energy demanded by the manufacturing industries. The low temperature (less than 100°C) corresponds to processes as washing, rinsing, and water and space heating of the industrial facilities, the medium temperature (100–400°C) corresponds to steam generation, and the high temperature (more than 400°C) corresponds to furnace and direct heat. The thermal energy use for processes

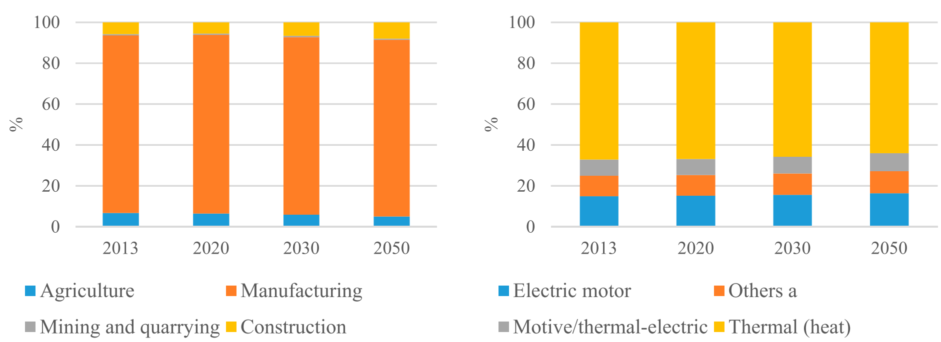

Figure 9. Share of sub-sector (left) and end-use (right) industry TFC of energy in the baseline scenario.
[a]Others include electricity for lighting, electric appliances, electro-chemical process, electro-thermal process and refrigeration.

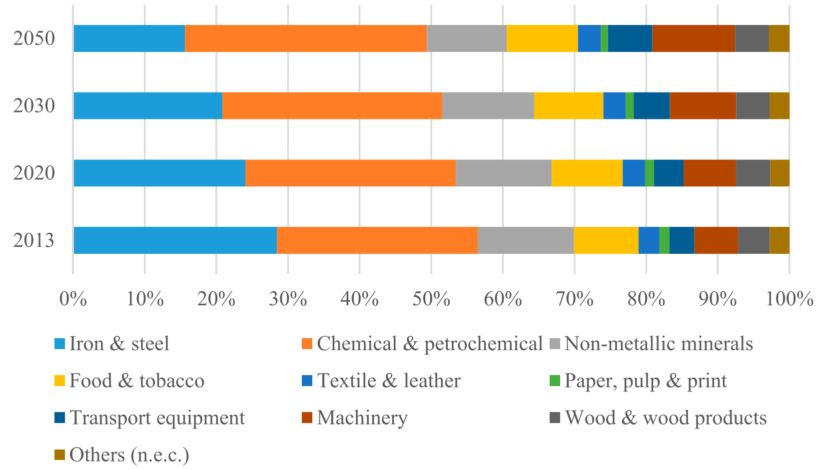

Figure 10. Industrial manufacturing sub-sector TFC of energy in the baseline scenario, 2013–2050.

such as washing, rinsing and space heating in agriculture, mining and quarrying, and construction sub-sectors, is relatively small. In 2050, the share of these three sub-sectors combined in total thermal energy demand is projected to be only 6.6%, about 449 ktoe in the baseline (Table 6). In the case of manufacturing industries, the demand for furnace and direct heat (high temperature) is projected increase from 2394 ktoe in 2013 to 2948 ktoe in 2050, an increase of 0.6% per year (Figure 11). However, its share in total thermal energy demand is projected to decrease from 56% in 2013 to 47% in 2050, mainly due to increase in the share of space and water heating (low temperature) in total thermal energy demand in the baseline.

5.2.4. Transport sector

Improvement of transportation remains very high on the policy agenda in Romania. In 2013, Romania's transport energy demand is 5278 ktoe or 24% of TFC of energy. Despite decrease in country's TFC over the past decade, energy use in transport is increasing steadily (1.3% per year), mainly due to a surge in passenger and freight road traffic. In 2013, 90% of Romania's transport energy demand

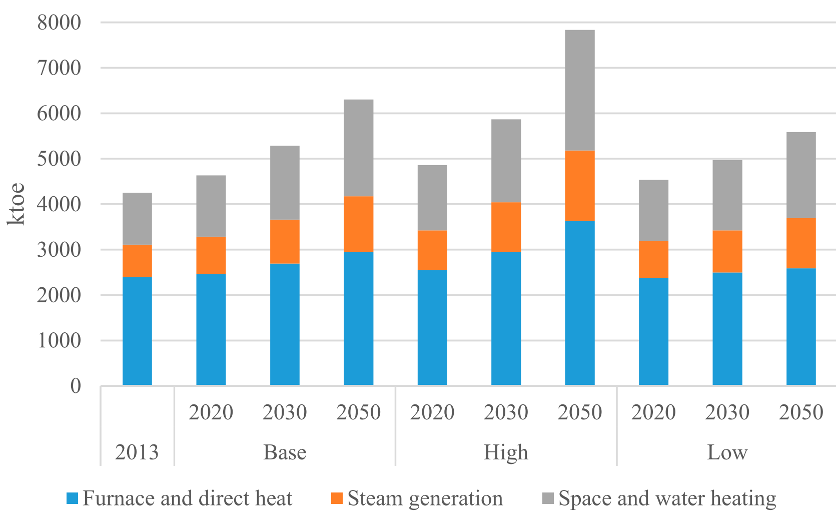

Figure 11. Industrial manufacturing TFC of non-electric (thermal) energy by scenario, 2013–2050.

Table 7. Transport sub-sector TFC of energy by scenario, 2013–2050 (ktoe).

		Baseline			High			Low		
	2013	2020	2030	2050	2020	2030	2050	2020	2030	2050
Air	211	215	232	313	218	240	336	214	228	301
Road	4766	5118	5508	6211	5189	5696	6684	5082	5416	5987
Rail	255	263	304	393	267	314	423	262	299	379
Inland waterways	42	41	44	71	41	46	77	40	43	69
Pipeline	4	4	4	6	4	4	6	4	4	5
Total (transport)	5278	5641	6092	6993	5720	6301	7526	5602	5990	6741

comes from road transportation and this share remains about the same over the projected period in all scenarios. The remaining transport energy demand is consumed mainly in air and rail transportation. Despite vehicle technology and fuel-economy improvements, transport energy demand is projected to increase during 2013–2050 period. Between 2013 and 2050, it is projected to increase by an average 0.8% per year to reach 6993 ktoe in 2050 in the baseline, compared to on average by 1% and 0.7% per year in high and low scenario, respectively (Table 7). Unless significant policy measures are adopted in promoting increasing use of mass transit system such as railways, road will remain the main transportation mode in the foreseeable future. As a consequence, this will also put pressure on demand for oil and associated negative air pollution impact.

Over the past decade, demand for passenger travel is also steadily increasing, while demand for freight travel is declined slightly in the country. Since 2004, the total passenger movement from road, rail and inland waterways combined is increased on average by 2.6% per year, while freight movement is decreased on average by 0.3% per year. The decline in country's freight movement over the past decade is mainly due to slump in economic growth in 2009 and 2010. However, as economy grows and the number of motorised vehicles increases, the passenger and freight travel demand is projected to increase over the 2013–2050 period. For example, excluding air and maritime transport, the passenger travel demand is projected to increase on average by 1.4% per year, reaching 174 giga-passenger-kilometre (Gpkm) in 2050, and the freight travel demand is projected to increase on average by 1.3% per year, reaching 98 giga-ton-kilometre (Gtkm) in 2050 in the baseline (Table 8). Compared to baseline, it follows the similar trends with higher travel demand in high scenario and with lower travel demand in the low scenario.

In Romania, road transport dominates both passenger and freight travel demand in the past and this trend is projected to continue during the projected period. For example, the road passenger travel demand is projected to grow on average by 1.4% per year, to reach 167 Gpkm in 2050 from 101 Gpkm in 2013 in the baseline (Table 8). More specifically, private passenger travel demand dominates road transport. Roughly 80% of road passenger travel demand is projected to be met by private

Table 8. Travel demand by transport mode by scenario, 2013–2050 (Gpkm, Gtkm)[a].

			Baseline			High			Low		
		2013	2020	2030	2050	2020	2030	2050	2020	2030	2050
Road	passenger public	20.7	21.6	23.6	30.8	22.7	24.8	32.3	20.7	22.7	29.5
	passenger private	80.4	82.4	89.9	135.9	86.5	94.4	142.7	79.1	86.3	130.4
	freight	34.0	38.8	44.1	57.5	40.7	46.3	60.3	37.2	42.4	55.2
Rail	passenger	4.4	4.8	5.4	6.9	5.1	5.7	7.2	4.6	5.2	6.6
	freight	12.9	12.4	13.6	18.4	13.0	14.3	19.3	11.9	13.0	17.6
Waterways	passenger	0.02	0.02	0.02	0.02	0.02	0.02	0.03	0.01	0.02	0.02
	freight	12.2	11.6	12.9	21.5	12.2	13.6	22.6	11.1	12.4	20.6
Pipeline	freight	0.8	0.9	1.0	1.0	0.9	1.0	1.1	0.9	0.9	1.0
Air	passenger	10.7	11.1	11.7	16.3	11.6	12.3	17.1	10.6	11.2	15.7
	freight	0.03	0.03	0.04	0.05	0.04	0.04	0.06	0.03	0.04	0.05
Maritime	passenger	0.05	0.06	0.07	0.08	0.06	0.07	0.09	0.06	0.06	0.08
	freight	43.6	52.0	62.7	83.6	54.6	65.8	87.8	49.9	60.2	80.3

[a]Passenger and freight travel demand for air and maritime are expressed in million passenger and million ton.

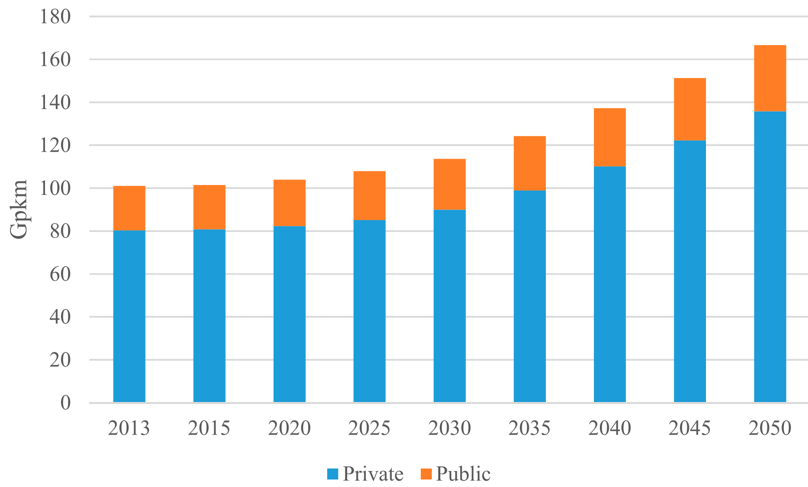

Figure 12. Road passenger transport travel demand in the baseline scenario.

passenger vehicles (mainly cars) in the baseline (Figure 12). The remaining 20% of total road passenger travel demand is projected to be met by public passenger vehicles (mainly buses and minibuses). The share of railways and inland waterways in total passenger travel demand is relatively small compared to passenger travel by road transport in the country. In 2050, the share of railways and waterways combined accounts to only 4% of total passenger travel demand in the baseline. In addition to lack of railway infrastructure, the projected decline in population and rising incomes over the study period will most likely limit the increase of passengers travel by railways.

Similar to passenger travel demand, freight travel demand is unevenly distributed among different transportation modes. In the freight transport, road remains the main transport mode during the projected period. For example, the share of road freight in total freight travel demand ranges from 57% in 2013 to 62% in 2030 and then decline slightly to 58% in 2050 (Figure 13). Road freight is followed by railways, inland waterways and pipeline during the projected period. These figures reflect, among others, the relative importance of passenger and freight transport in the country.

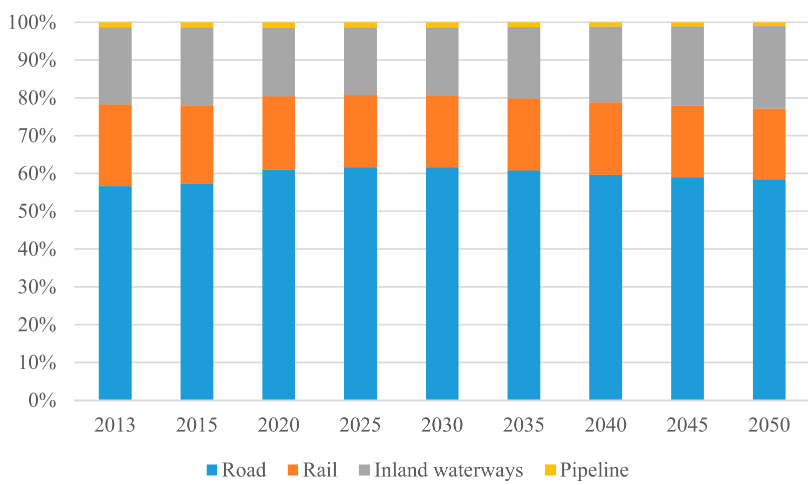

Figure 13. Share of freight transport demand by transport model in the baseline scenario.

While passenger travel patterns are more closely related to personal wealth and lifestyle changes, freight transport activities are closely connected to overall economic activity.

In recent years, both air passenger volumes, measured in number of passengers, and air freight volumes, measured in ton, grew steadily, closely connected to country's economic growth rate. It is projected that the trend will continue over the projected period. For example, air passenger volumes is projected to grow at an average rate of 1.1% per year during 2013–2050 to 17 million passengers in 2050, while air freight volumes is projected to grow at 1.4% per year over the same period to 55000 ton in 2050 (Table 8). Maritime follows similar trend as that of air transport during the projected period.

6. Concluding remarks

In recent years, Romania's economy is growing at much faster rate than most of EU countries, while country's energy demand growth remained flat. This decoupling of energy use and economic growth is mainly due to decline of energy-intensive industries and growth of the service sectors. This study delivers a quantitative assessment of sector-wise energy demand by end-uses and travel demand by transportation modes over the next 35 years. Unlike in the past, the country's overall TFC of energy is expected to increase steadily in the future. In particular, demand for electricity will continue to grow. Despite this, the gap in per capita electricity consumption in Romania and EU member countries will likely remain very large in the future.

Residential sector is presently the largest energy consumer, followed by industry, transport and services sector. However, growth in energy demand across the sectors varies widely with time. The growth in residential TFC of energy is projected to slow with time, mainly due in part to declining population and using more efficient household appliances. Most of residential energy demand comes from space and water heating activities. Residential space heating, therefore, represents the largest opportunity to reduce residential energy demand in the country by modernising district heating systems, including energy efficiency improvements of the heating equipment, and by changing energy-mix. From 2035 onwards, industry will surpass residential sector in TFC, mainly due to strong growth in energy-intensive manufacturing industries. Although the share of services in TFC is relatively small, this sector's TFC of energy will double by 2050. Almost half of total services energy demand will come from space heating in office buildings, hotels, and wholesale and retail stores. Besides using improved and efficient heating equipment, there is also great potential in reducing energy demand by renovating office buildings. Transport, mainly road, is another important sector where steady increase in energy demand is expected. Although reducing energy demand in the transport sector is a particularly difficult challenge in the country, Romanian government is encouraged to intensify its efforts to shift from private to public road and rail transport by promoting mass transit system.

We acknowledge that one important limitation of this study is lack of rural-urban and regional differences of residential energy demand analysis. If reliable data become available, further works that explore these issues would be worthwhile. We also note that our model focuses on energy demand and provide limited information on energy supply system. Understanding of these energy demand and supply evolution is important in implementing EU recommended energy and climate strategies in the country would be interesting for further study.

Notes

1. Different modeling techniques include mathematical programming, econometrics, statistical and accounting methods, and network analysis. The Long-range Energy Alternatives Planning system (LEAP), Asia-Pacific Integrated Model (AIM)/Enduse, TIMES/MARKAL, National Energy Modeling System (NEMS), ODYSSEE and Model for Analysis of Energy Demand (MAED) are some of the commonly used end-use energy demand models. See Capros et al. (2014) for the description of seven large-scale energy-economy EU models (PRIMES, GEM-E3, TIMES-PanEu, NEMESIS, WorldScan, Green-X and GAINS) used to assess European

decarbonisation pathways. Likewise, Novosel et al. (2015) use agent-based tool (MATSim) and EnergyPLAN for transport energy demand modeling. For industrial energy demand models, see Edelenbosch et al. (2017).
2. The EEDA model development is part of the 'Romania: Climate Change and Low Carbon Green Growth Program' conducted by Ministry of Environment and Climate Change of Romania (now, Ministry of Environment, Water and Forest) and the World Bank (World Bank 2015). The model is built using MS Excel and its most important features include transparency, flexibility and easy to use.
3. Most of the figures and tables in Section 5 are similar to those that are presented in our working paper version.
4. These studies include EU's energy, transport and GHG emissions trends to 2050 (EU 2016), Romania's Ministry of Environment and Climate Change's sixth national commutation (MECC 2013)] and the National Commission for Prognosis (NCP)'s energy balance forecast (NCP 2015).
5. Despite declining population during the 2013–2050 period, demand for heating energy is projected to increase mainly due to increase in size of living floor space in the country. For example, between 2013 and 2050, living floor space is projected to increase on average by 0.3% per year.

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Disclosure statement

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Appendices

Appendix 1. Key driving variables under the baseline, high demand and low demand scenarios for selected years (Index 2013 = 100, unless otherwise stated)[a]

	Base case			High			Low		
	2020	2030	2050	2020	2030	2050	2020	2030	2050
Population[b]	99	95	90	99	96	91	98	95	87
GDP[c]	127	163	233	133	171	244	122	156	223
Residential effective floor space[d]	103	107	113	105	112	124	102	105	108
Appliances ownership[e]									
Refrigerator/freezer	108	118	140	111	128	160	104	109	120
Washing machine	105	112	126	105	112	126	105	112	126
Television	107	118	138	107	118	138	107	118	138
Computer	142	204	327	142	204	327	142	204	327
Others	112	129	163	112	129	163	112	129	163
No. of residential dwellings with AC[f]	135	189	285	135	190	290	135	187	275
Services buildings floor space[g]									
Offices (public and private)	119	143	192	124	156	232	115	131	160
Educational buildings	113	132	163	117	144	197	108	120	135
Hospitals	113	132	163	117	144	197	108	120	135
Hotels and restaurants	126	155	227	131	169	273	122	142	188
Sport facilities	113	132	163	117	144	197	108	120	135
Wholesale and retail trade	126	155	227	131	169	273	122	142	188
Others (n.e.c.)[h]	119	143	192	124	156	232	115	131	160
Services buildings demolition rate (%)	1.3	2.3	1.0	1.3	2.3	1.0	1.3	2.3	1.0
Services buildings occupancy rate (%)	85	85	85	85	85	85	85	85	85
Industrial domestic production[b]									
Agriculture	107	119	130	111	129	156	103	109	109
Manufacturing									
Iron and steel	98	104	107	99	110	125	92	93	89
Chemical and petrochemical	127	170	279	140	200	371	131	170	267
Non-metallic minerals	115	136	162	115	142	188	107	120	133
Food and tobacco	122	143	189	128	158	231	120	134	165
Textile and leather	122	143	189	128	158	231	120	134	165
Paper, pulp and print	100	107	122	102	115	145	95	97	105
Transport equipment	134	196	311	139	212	371	130	180	261
Machinery	136	204	330	140	221	394	131	187	278
Wood and wood products	122	143	189	128	158	231	120	134	165
Other manufacturing (n.e.c.)	111	135	180	115	140	208	107	118	145
Mining and quarrying	102	111	129	105	119	147	98	101	102
Construction	106	152	235	108	162	278	101	137	197
Transport domestic demand[c]									
Transport (Air)	109	127	201	115	134	211	105	122	193
Transport (Road – Freight)	143	193	286	150	203	301	137	185	275
Transport (Road – Passenger)	112	142	222	118	149	234	108	136	214
Transport (Rail – Freight)	102	121	194	107	127	204	98	116	186
Transport (Rail – Passenger)	116	147	218	122	154	229	111	141	210
Transport (Water – Freight)	97	111	194	102	116	204	93	106	187
Transport (Water – Passenger)	98	113	208	103	119	218	94	108	199
Gas pipeline	112	126	142	117	132	149	107	121	136
Rebound effect[i]	103	107	115	103	107	115	103	107	115

[a]This research study is part of 'Romania: Climate Change and Low Carbon Green Growth Program' conducted by Ministry of Environment and Climate Change of Romania (now, Ministry of Environment, Water and Forest) and the World Bank (World Bank 2015). The assumptions related to key future driving variables are based on World Bank's ROM-E3 model and authors' ESDA model.

[b]Based on Eurostat's main variant (baseline), high life expectancy variant (high) and low fertility variant (low) population projection (Eurostat 2016d).

[c]GDP, industrial domestic production and transport demand projections are based on World Bank's ROM-E3 simulation model data (see World Bank 2015).

[d]Residential effective floor space projection is calculated using dwelling occupancy rate, dwelling stock, demolition rate dwelling area, population growth and household size based on ESDA model.

[e]Appliances ownership are based on ESDA model.

[f]The share of dwelling with air conditioning (AC) is assumed to grow by 2% in every five years over the projected period. This value is 5% in the base year.

[g]Calculation of services floor space projection follows same approach that of residential sector using ROM-E3 simulation model data of domestic demand of market and non-market services.

[h]The abbreviation n.e.c. is not elsewhere classified.

[i]Rebound effect (positive) is considered mainly due to income and fuel-economy improvements for passenger road transportation only. The figures for this effect are authors' assumption and they are same for all three scenarios.

Appendix 2. Sector/sub-sector energy intensity by end-use under the baseline scenario for selected years

Sector/sub-sector	Unit[a]	Base year 2013	Index (2013 = 100) 2020	2030	2050
Residential					
Space heating	kWh/sqm	319	100	100	100
AC	kWh/dw	248	100	100	100
Water heating	kWh/cap	202	100	100	100
Cooking	kWh/cap	76	100	100	100
Lighting	lm/sqm	200	100	100	100
Electric appliances	kWh/dw				
Refrigerator/freezer		125	98	95	94
Washing machine		795	99	98	97
Television		403	100	99	98
Computer		245	99	98	97
Others		150	100	99	99
Services	kWh/sqm				
Space heating					
Offices		176	100	100	100
Educational building		42	100	100	100
Hospitals		42	100	100	100
Hotels and restaurants		205	100	100	100
Sport facilities		98	100	100	100
Wholesale and retail trade		123	100	100	100
Other non-residential buildings		38	100	100	100
Space cooling					
Offices		34	100	100	100
Educational building		9	100	100	100
Hospitals		9	100	100	100
Hotels and restaurants		53	100	100	100
Sport facilities		18	100	100	100
Wholesale and retail trade		15	100	100	100
Other non-residential buildings		5	100	100	100
Offices		34	100	100	100
Educational building		8	100	100	100
Hospitals		8	100	100	100
Hotels and restaurants		49	100	100	100
Sport facilities		18	100	100	100
Wholesale and retail trade		16	100	100	100
Other non-residential buildings		8	100	100	100
Lighting[b]		400	100	100	100
Others		56	100	100	100
Industry	Wh/€				
Agriculture		31	98	96	92
Mining and Quarrying		3	98	96	92
Construction		19	98	96	92
Manufacturing					
Iron and steel		3411	95	90	82
Chemical and petrochemical		1170	90	81	66
Non-metallic minerals		1803	95	90	82
Food and tobacco		150	98	96	92
Textile and leather		107	98	96	92
Paper, pulp and print		188	98	96	92
Transport equipment		37	98	96	92
Machinery		109	98	96	92
Wood and wood products		230	98	96	92
Transportation					
Air					
Passenger	passenger/1000 €	10.7	95	87	77
Freight	ton/1000 €	0.03	97	92	85
Road					

(*Continued*)

Continued.

Sector/sub-sector	Unit[a]	Base year 2013	Index (2013 = 100)		
			2020	2030	2050
Passenger public	pkm/1000 €	20675	94	81	67
Passenger private	pkm/1000 €	80,363	92	80	77
Freight	tkm/1000 €	34,026	84	70	62
Rail					
Passenger	pkm/1000 €	4411	95	84	72
Freight	tkm/1000 €	12,941	93	86	72
Inland waterway					
Passenger	pkm/1000 €	17	94	84	72
Freight	tkm/1000 €	12,242	99	97	92
Maritime					
Passenger	passenger/b€	0.05	95	87	77
Freight	ton/b€	44	97	92	85
Pipeline freight	tkm/1000 €	829	97	92	85

[a]kWh is kilo-watt hour, sqm is square metre, dw is dwelling, cap is capita, pkm is passenger-kilometre, tkm is ton-kilometre, b€ is billion euro.
[b]Lighting unit is lumen/sqm.
Sources: Atanasiu, Economidou, and Maio (2012); Pardo et al. (2012); ITU (2014); MECC (2013); BPIE (2014); World Bank (2015); INS (2016a, 2016d); Eurostat (2016a, 2016b, 2016c).